

Goal. Evaluate integrals by partial fractions and/or other methods

1. Evaluate $\int \frac{x+8}{x^2+x-2} dx$ using the steps for partial fractions.

2. Suppose you wanted to evaluate these integrals using partial fractions. What is the form of partial fractions you would use?

(a) $\int \frac{x}{(2x-1)^4} dx$

(b) $\int \frac{22x+4}{(x-3)^2(x^2+1)} dx$

3. Evaluate $\int \frac{(5x^3 + 1)(x^2 + x)}{(x^3 - x)(x - 1)} dx$. First, be sure to factor the denominator fully and do any algebraic simplification.

4. Evaluate integral 2(a) by a method other than partial fractions. What method did you use?

5. Compute the following integral

$$\int \frac{dx}{x^2 + 6x + 8}$$

6. Use partial fractions to write integral 2(b) as the sum of several simpler integrals. Decide which of those integrals you can integrate. Integrate the ones you can, and explain why the methods you currently know fail to integrate the others.